SAR TECHNICAL GIS & REMOTE SENSING EXPERT PACKET

Bare-Earth LiDAR Differentials, Cellular Telemetry, Behavioral Decay Functions, and Anomaly Registers

1. Theoretical SAR Behavioral Modeling & Distance Decay

The spatial analysis incorporates behavioral data models from **Syrotuck (1975)**, **Koester International Lost Person Behavior (2008)**, **Colwell Probability of Detection (POD) sweep configurations**, and **O'Brien et al. (2021)** ordinal regression models on 16,454 missing person incidents.

- **Downhill Travel Vectors:** In alpine/sub-alpine terrain, 56% to 86% of lost subjects descend along gravity-driven contours, making drainage channels primary high-probability density zones.
- **Distance Decay Functions:** Ordinal regression models confirm that adult males lacking planning indicators exhibit tight distance decay, with 69.3% located within 0–5 miles (median 1.0–2.6 km).
- **Missing 411 / GSAR Profile Points:** Inability of trailing K9s to acquire scent across frozen snowpack, abrupt blizzard onset obscuring tracks, and proximity to steep drainage/water features.

2. Cellular Telemetry & RF Disconnection Forensic Protocol

To determine whether cell phone communication ceased due to battery depletion, physical destruction, or terrain dead-zone masking, RF engineers analyze the VLR/HLR network event logs:

Disconnect Scenario	Protocol Signaling	Network Timeout Behavior	Investigative Interpretation
Graceful Shutdown	IMSI Detach / Deregistration	Immediate clean socket closure	Manual power-off or drained battery
Sudden Physical Destruction	No Detach Packet Sent	Server-side paging timeout (2-5 min)	Catastrophic impact, crushing, or water immersion
Topographic Dead-Zone	Gradual RSSI/RSRP Degradation	Handshake timeout during motion	Vehicle or foot entry into mountain ravine/cut

3. Technical Target Registers (Mont-Tremblant & Sun Peaks)

Target ID	Name & Location	Category	GPS Coordinates	Void Depth	Concealment Score
LT-HAZARD-01	Pied-de-la-Montagne Culvert	Drainage Culvert / Trench	46.2098° N, -74.5848° W	1.40 m	0.92
LT-HAZARD-02	Voyageurs II Parking Lot	Snowbank Retention Void	46.2092° N, -74.5855° W	0.95 m	0.88
LT-HAZARD-03	Lac Miroir Outflow Creek	Stream Cut-Bank / Tree Well	46.2085° N, -74.5861° W	1.75 m	0.95
LT-HAZARD-04	Decommissioned Utility Vault	Subsurface Concrete Basin	46.2102° N, -74.5842° W	2.10 m	0.89
LT-HAZARD-05	South Pedestrian Path	Conifer Tree Well Cluster	46.2089° N, -74.5838° W	1.50 m	0.85
RS-TARGET-DELTA	Fairways Creek Ravine	Bare-Earth LiDAR Sink	50.8782° N, -119.9075° W	2.40 m	0.96